Week 12 web page: delivery guide

Instructor notes for Week12_MixedModels.html

General

- Page: https://kevintshoemaker.github.io/NRES-746/Week12_MixedModels.html. Script: `CODE_Week12_MixedModels.R` (same seeds, same numbers; run its first section before any later one).
- Veterans Day: no class Wed., Nov. 11. The week is Monday plus Tuesday's lab period.
- Monday has **no WOD** (the lecture is full). The required take-home WOD *Formula to Model* goes out Monday, due Mon., Nov. 16; budget 5-10 min that day to go over it. Key: `wod/answer_keys/`.
- Tuesday (revised Nov. 9): ~60 min of lecture finishing Part 1 (random slopes, mixed models in R, discussion), then an open final-project work session. No lab Act or key. Part 2 (Bayesian hierarchical models) moves to Mon., Nov. 16.
- Two simulated data sets; say so up front.
  - Pinyon chronologies (seed 21; 4 stands × 5 trees × 25 yr): truth stand SD 0.12, tree SD 0.10, precipitation slope 0.025 mm/cm, AR(1) $\phi = 0.6$, **no trend**.
  - Drought sensitivity (seed 12): 12 stands, records of 30, 24, 18, 12, 8, 5 years (two stands each), 194 stand-years. Precipitation in 10-cm units centered at 20 cm. Truth: $\mu_a = 0.9$, $\mu_b = 0.25$, $\sigma_a = 0.15$, $\sigma_b = 0.10$, $\rho_{ab} = -0.6$, $\sigma_e = 0.12$.
- Questions on the page have no answers. *Land:* is where discussion should end up.

Part 1: Autocorrelation and mixed models in R (Monday, ~45 min)

Sections 1-2 are the Week 11 material that wasn't reached on Nov. 4; the Week 11 page now points here.

As taught (Mon., Nov. 9): the opening recap (mean vector and V; why the MVN; GLMMs as a latent normal layer), a review of the random intercept's block-diagonal V, AR(1) (most of the period: building V, contrast with the block-diagonal case, the widget, total vs. innovation variance), and why these models are harder to fit. **Not covered, moved to Tuesday's first hour:** random slopes, mixed models in R, and the discussion. The lagged-response subsection and the bonuses were not covered either. Part 2 moves to Mon., Nov. 16.

As taught (Tue., Nov. 10): the first hour of the lab period covered the rest of Part 1,

everything on the page up to the Bayesian H1 (`#bayesian-hierarchical`). The rest of the period was the final-project work session. Part 2 is unchanged and stays on Mon., Nov. 16.

Big picture (~3 min)

- MODELS and FIT boxes. Recap in one breath: random intercept → block-diagonal V. Today: correlation through time, then groups that differ in slope, then the software.

AR(1) (~10 min)

- Management question first: is growth declining? `lm()`: trend -0.0055 mm/yr, SE 0.0026, $p = 0.045$. Stand A's lag-1 residual ACF (0.33) is inside the ± 0.39 bounds: not detected $\neq$ absent. All four stands are positive (0.33, 0.33, 0.48, 0.41).
- Derive $\mathrm{Var}(\eta_t) = \sigma_e^2/(1-\phi^2)$ on the board (two lines). Notation follows Hyndman & Athanasopoulos: η_t is the autocorrelated error, ε_t the independent innovation; `nlme` prints ϕ as `Phi`. AR(1) fit: $\hat{\phi} = 0.37$; trend -0.0047, SE 0.0034, $p = 0.17$. `gls()` matches. Then tell them the truth has no trend.
- Widget (`cov_structures.html`): switch to AR(1), default ϕ 0.6 (lag-1 0.60, lag-5 0.08). Ask students to predict the series at $\phi = 0.9$ before moving the slider: it wanders like a trend (lag-5 still 0.59). Then $\phi = -0.7$: zigzag, checkerboard heatmap.
- "With ϕ near 0.9, what does a single AR(1) series look like? How could you tell it apart from a slow trend?" *Land:* long runs above and below the mean that look like trends. From one short series you often can't; you need a longer record, replicate series, or a mechanism. This is the fan in section 2.
- *If time:* AR errors vs. lagged response (`#lagged-response`). Board the substitution. Stand A: plain lagged fit $\hat{\gamma} = 0.02$ (truth $\phi = 0.6$); with last year's precip $\hat{\gamma} = 0.33$ and lagged precip -0.0089 vs. $-\hat{\gamma}\hat{\beta} = -0.0070$. Point: choose the form by mechanism (does the response itself carry over?).

Why harder (~7 min)

- Three points, one line each: cost (about 150 ms per MVN evaluation at $n = 2000$ on the render machine), awkward surfaces, constraints.
- Random-intercept surface: σ_a about 0.08 to 0.35; σ_e much tighter.
- Fan: near $\phi = 0$ the trend runs about -0.0074 to -0.0039 (excludes 0); at $\hat{\phi} = 0.37$ it runs -0.0113 to 0.0017.
- **Behind?** Skip the timing; show the two surfaces.

Tuesday, Nov. 10 (first hour of the lab period) starts here. Open with 2 min recapping AR(1) and the block-diagonal V; with ~60 min there's room for the discussion and one or two bonuses (lagged response, GLS).

Random slopes (~8 min)

- Show the two-panel figure. Right panel: low-intercept stands have steep slopes. Ask what ecology would produce that (low elevation, closer to the water limit).
- Write the model and G on the board; count the six parameters. The 24 stand deviations are integrated out, not estimated.

- $V_j = Z_j G Z_j^T + \sigma_e^2 I$: one block per stand. Don't expand Var(y) as a function of x (the take-home WOD asks for it).
- Formula table: `(1 | stand/tree)` expands to `(1 | stand) + (1 | stand:tree)`. Don't count parameters for them out loud beyond the table (the WOD uses different formulas).

Mixed models in R (~10 min)

- `lmer()`: $\hat{\mu}_a = 0.840$, $\hat{\mu}_b = 0.258$, $\hat{\sigma}_a = 0.131$, $\hat{\sigma}_b = 0.092$, $\hat{\rho}_{ab} = -0.57$, $\hat{\sigma}_e = 0.118$. AIC -180.9 (random intercept) vs. -216.9 (random slope), a drop of 36.0.
- If asked about an LRT: $\chi^2 = 40.0$ on 2 df, $p \approx 2 \times 10^{-9}$; conservative because $\sigma_b = 0$ is on the boundary (same issue as Poisson vs. negative binomial).
- The comparison table: `optim()`, `lmer()`, `lme()`, and `glmmTMB()` all give log-likelihood 114.4497. The point: the packages fit the likelihood we could write ourselves, faster and with better parameterizations.
- Package bullets: glmmTMB as the practical default (one syntax for non-Normal families, dispersion, AR(1) and spatial structures); lme4 for small-sample inference on Normal models (REML, `lmerTest`, profile intervals); nlme for the richest Normal residual structures. Point students to the reference handout `NOTES_Reference_MixedModelSyntax.pdf` (linked on the page).
- **Behind?** Show only the `lmer()` fits and the comparison table; skip the `optim()` code walk-through.

Discussion (~5 min; pick one or two)

- Q1 (trend SE 0.0026 to 0.0034; precipitation SE 0.0026 to 0.0020) *Land:* year is itself perfectly autocorrelated, so runs of correlated errors line up with it and mimic a trend; precipitation jumps around, so correlated errors don't line up with it. Errors matter most for predictors that change slowly in time or space.
- Q2 (four stands: fixed or random?) *Land:* the random-intercept model answers "how variable are stands in general" and can predict a new stand, but σ_a from four stands is weak (wide surface). The fixed model answers "how do these four stands differ." Choose by the question; with few groups, say so and report the uncertainty.
- Q3 (four short-record stands: $\hat{\sigma}_a = 0.084$, $\hat{\sigma}_b = 0.063$, $\hat{\rho}_{ab} = -1.00$, singular fit) *Land:* no; 26 stand-years in 4 stands can't support three variance parameters, and $\hat{\rho}_{ab} = -1$ is the boundary (the fit collapsed to one dimension). Report it as singular; simplify to `(precip10 || stand)` or a random intercept, get more stands, or use a prior (Part 2).

Bonuses (only if time): AR(1) by GLS (closed-form $\hat{\beta}(\phi)$ and $\hat{\sigma}_e^2(\phi)$, then `optimize()` over ϕ; matches `optim()` and `gls()`, $\hat{\phi} = 0.367$, logLik 27.037; profile interval for ϕ -0.03 to 0.75, so neither 0 nor the true 0.6 is ruled out; `gls()` reports $\sigma_e/\sqrt{1-\phi^2}$, not the innovation SD); random intercept + AR(1) ($\hat{\phi} = 0.51$, log-likelihood 82.28, matches `lme()`); Poisson GLMM by hand (`integrate()`: 1.551, 0.382, σ_a 0.542; `glmmTMB()` 1.548, 0.382, 0.542).

Part 2: Bayesian hierarchical models

Moved Nov. 10 to the Week 13 page and to Part 1 of `NOTES_Week13_WebPageGuide.pdf` (Mon., Nov. 16).